University of Florida, Department of Mathematics

Probability PhD Exam, May 2003

Write clearly. State precisely results you use. Do not omit steps.

1. Let $(\overline{X}_m)_{m=1}^{\infty}$ be an i.i.d. sequence of real valued variables on $(\Omega, \mathcal{F}, P)$. Let $\mathcal{F}_m = \sigma(\overline{X}_1, \dots, \overline{X}_m)$. Put $S_m = \overline{X}_1 + \cdots + \overline{X}_m$, for $m \geq 1$, and put $S_0 = 0$. Which of the following sequences are martingales relative to $\mathcal{F}_m$? For those which are not martingales with respect to this filtration, what conditions have to be added to make them martingales?

a) S_m, $m = 0, 1, 2, \dots$, $E[|\overline{X}_1|] < \infty$.

b) $\overline{X}_1^2 + \cdots + \overline{X}_m^2 - m\lambda$, $m = 1, 2, \dots$, $E(\overline{X}_1^2) < \infty$, λ real.

c) $\exp(\alpha S_m - m\lambda)$, $m = 0, 1, 2, \dots$, where $\varphi(\alpha) = E[\exp(\alpha \overline{X}_1)] < \infty$, α, λ real.

d) $\overline{Y}_m = S_{\min(m,T)}$, $m = 0, 1, 2, \dots$, where $P(\overline{X}_1 = \pm 1) = \frac{1}{2}$ and $T = \min\{n > 0 : S_n = 0\}$.

2. Let $\{\alpha_m\}_{m=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$. Show the following are equivalent:

a) There exists a probability measure α on $\mathbb{R}$ such that $\lim_{m\to\infty} \alpha_m(I) = \alpha(I)$ for all closed intervals I on $\mathbb{R}$ whose end points are continuity points of α.

b) If $\{F_m\}$, respectively F, are the distribution functions of $\{\alpha_m\}$, respectively α, then $\lim_{m\to\infty} F_m(x) = F(x)$ at every point x of continuity of F.

3. Let $\{\alpha_m\}_{m=1,2,\dots}$ be a sequence of probability measures on $\mathbb{R}$, and let $\{\varphi_m(t)\}_{m=1,2,\dots}$ be the corresponding characteristic functions. Assume that $\lim_{m\to\infty} \varphi_m(t) = \varphi(t)$ for all $t \in \mathbb{R}$, and $\varphi(t)$ is continuous at $t = 0$. Prove that $\varphi(t)$ is the characteristic function of some probability measure and that the conditions of Problem 2 hold.

4. If $\overline{X}$ is a real valued random variable, show that for $0 < \varepsilon < 1$,

$$E\left[\frac{|\overline{X}|}{1 + |\overline{X}|}\right] \leq \varepsilon + P[|\overline{X}| > \varepsilon]$$

and that

$$P(|\overline{X}| > \varepsilon) \leq \frac{1+\varepsilon}{\varepsilon} E\left[\frac{|\overline{X}|}{1 + |\overline{X}|}\right].$$

Deduce that the set of real valued random variables on a given probability space can (with suitable identification) be regarded as a metric space if the distance between two random variables $\overline{X}$ and $\overline{Y}$ is defined to be

$$d(\overline{X}, \overline{Y}) = E\left[\frac{|\overline{X} - \overline{Y}|}{1 + |\overline{X} - \overline{Y}|}\right],$$

and that convergence in this metric space corresponds to convergence in probability.

5. This problem has three parts.

a) Let $\overline{X}_m$, $m = 1, 2, \dots$, be i.i.d., $\overline{X}_m > 0$, $E[\overline{X}_m] = 1$, and $\overline{Y}_m = \prod_{k=1}^m \overline{X}_k$. Prove that $\overline{Y}_m$ is a martingale relative to the filtration $\mathcal{F}_m = \sigma(\overline{X}_1, \dots, \overline{X}_m)$. Show that $\lim_{m\to\infty} \overline{Y}_m = \overline{Y}_\infty$ exists a.s.

b) Show that $P[\overline{Y}_\infty = 0] = 0$ or 1.
Hint: Recall infinite products and the zero-one law.

c) Using b), show that

$$E[\overline{Y}_\infty] = 1 \quad \Longleftrightarrow \quad P[\overline{Y}_\infty > 0] = 1.$$

6. Let $\overline{X}_m$, $m = 1, 2, ...$, be a supermartingale such that

$$E[|\overline{X}_{m+1} - \overline{X}_m| \mid \mathcal{F}_m] < M \quad \text{a.s.}$$

Show that for any two optional times $S \le T$, $E[T] < \infty$, we have $E[|\overline{X}_T|] < \infty$ and

$$E(\overline{X}_T \mid \mathcal{F}_S) \le \overline{X}_S.$$